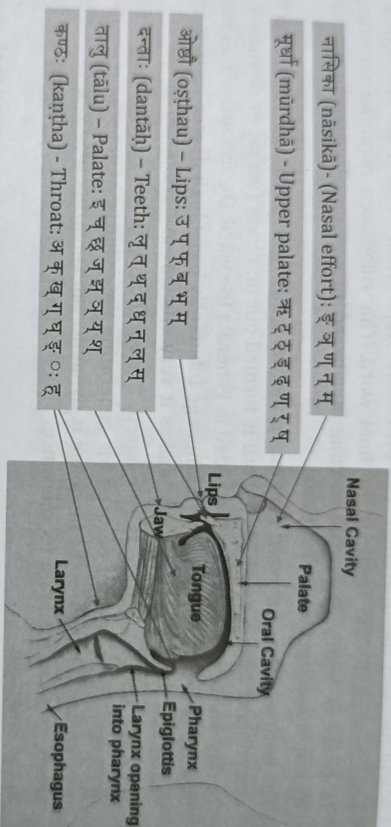


PHONETICS IN SANSKRIT

Phonetics is the study of sounds in a language, particularly the production of sound in a language and how it communicates the language corresponding to the scripts of the language. It also addresses the issue of how the sound is perceived in the language. Phonetics in the Sanskrit language has been addressed in some detail, since this is vital, because the ancient Indian knowledge tradition is oral. The entire transmission of the Vedas from time immemorial has survived several thousand years on account of scientific methods of oral rendering. This has been possible on account of a well-developed science of phonetics. This also ensured an impeccable textual transmission superior to the classical texts of other cultures. This is perhaps the reason for UNESCO recognizing Vedas in the form of oral knowledge as a heritage for preservation.

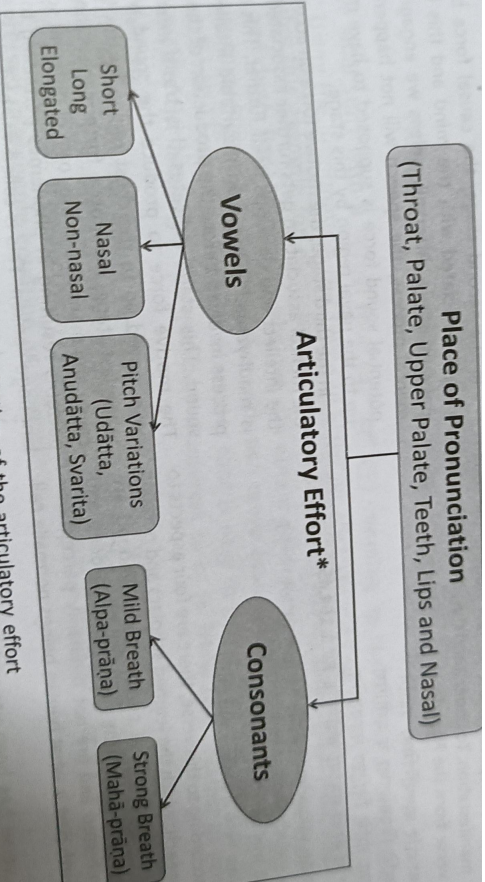
According to Panini, the vowels have a temporal factor in the production of the sound! Three variations have been specified; short (hrasva), long (dirgha), and prolate or elongated (pluta). If we take the duration of the utterance of a short vowel as one unit, the long will be two units and the prolate Vedos from time immemorial will be three units. One can represent them using the letters has been possible on account as अ, आ, ॐ

Asādhyay! establishes that in our effort to utter a vowel, it is possible to use either the oral cavity alone or the nasal cavity along vowel sound. with the oral cavity: When a vowel is uttered by the joint effort of the oral and the nasal cavities, it is referred to as nasal (anunāsika). By contrast, a vowel uttered only using the oral cavity may be referred to as non-nasal (nir-anunāsika). The third aspect of the vowel sound is the pitch used to utter the vowel. Three variations have been specified for the pitch of the sound: A vowel uttered with high pitch is called udatta, with low pitch anudatta and that with a combination of high and low pitches known svarita.



Source: https://commons.wikimedia.org/wiki/File:llu01_head_neck.jpg

FIGURE 5.2 Origin of Sound from the Oral Cavity – An Illustration



* This presents only a partial categorisation of the articulatory effort

FIGURE 5.3 Place of Pronunciation and Articulatory Effort in Sanskrit